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Letter of Recommendation for Dr. Jialun Cao

Dear Committee Members,

I am delighted to provide a letter of reference for Dr. Jialun Cao's application at your institution, and I recommend considering her candidacy in the **strongest possible terms**. Jialun is an outstanding researcher who represents exactly the kind of talent that top academic institutions should actively seek to recruit. I clearly consider her to be a rising star in software engineering, and I am confident that she will make any academic institution, including the most prestigious ones, substantially stronger. Her exceptional promise is evidenced by her receipt of the **ACM SIGSOFT Outstanding Doctoral Dissertation Award**, an honor given only once per year to recognize the best dissertation in software engineering worldwide. It is noteworthy that practically all past winners of this award have gone on to highly successful careers, predominantly at top-tier academic institutions, a trajectory I fully expect Jialun to follow.

Before elaborating on Jialun's qualifications, I would like to provide some context regarding my interactions with her. I first met Jialun at FSE 2023, where she presented part of her PhD work on testing NLP models, and I was deeply impressed by the quality of her research and her presentation skills. In 2024/2025, I had the pleasure of hosting Jialun as a visiting scholar at the University of Stuttgart. This experience led to several ongoing collaborations, allowing us to meet regularly to discuss our work together. Through these interactions, I gained a close-up view of her remarkable research skills and her admirable personality.

Based on my interactions with Jialun, I can confidently state that she possesses the technical leadership and skill set necessary to excel in a tenure-track position at a top-tier institution. She has the ability to identify relevant research problems, define exciting research ideas to address them effectively, and critically evaluate their impact. Furthermore, Jialun demonstrates strong supervision skills, having mentored MSc-level and PhD-level students for several years. She is a motivated, hard-working, and persistent researcher who consistently strives for excellence in her work.

Dr. Jialun Cao works at the intersection of software engineering and artificial intelligence, a topic of great relevance and impact today and in the foreseeable future. Her focus on improving software testing and software verification through the use and development of novel machine learning techniques positions her as a leader in this field. Jialun brings the right mix of skills and knowledge gathered from her prior work to make a strong impact in this area, as she has an outstanding background in software engineering and has also made significant contributions to software verification and machine learning. Her **impressive breadth of topics** includes AI-supported techniques for enhancing the **entire software development lifecycle**, encompassing requirements, code completion, bug detection, test generation, program repair, and formal verification.

Jialun's perhaps most impressive line of work is on **AI-assisted formal verification**, which is a **new research direction** that she has pioneered and advanced significantly, as exemplified by her ACL'25 paper "From Informal to Formal – Incorporating and Evaluating LLMs on Natural Language Requirements to Verifiable Formal Proofs" and her CAV'24 paper "Enchanting Program Specification Synthesis by Large Language Models Using Static Analysis and Program Verification". Together, these two papers articulate a coherent and forward-looking research agenda aimed at bridging the long-standing gap between informal

program intent and fully verifiable formal artifacts by leveraging large language models in a principled, verification-centric way. The first paper establishes a foundational understanding of this space by decomposing the informal-to-formal pipeline into fine-grained tasks, constructing large-scale multi-language benchmarks, and providing rigorous empirical evidence on where current LLMs succeed, fail, and improve through targeted fine-tuning. Building on this understanding, the second paper advances the state of the art by operationalizing LLMs within a sound verification loop, where static analysis and theorem provers guide, validate, and refine specification synthesis in an iterative and scalable manner. Taken together, these papers demonstrate how **LLMs can be transformed from brittle generators into reliable collaborators in formal verification**. This line of research significantly lowers the barrier to rigorous software verification and sets a compelling blueprint for future systems that tightly integrate learning-based models with symbolic reasoning and formal guarantees. Given the increasing use of LLMs in software engineering, yet the persistent challenges around correctness and reliability, this line of Jialun's research promises to be the beginning of an impactful and influential research trajectory.

Dr. Jialun Cao is not only an exceptional researcher but also a highly engaged member of the academic community. Given her highly visible and impactful research contributions, she is frequently invited to service roles within the research community. I witnessed first-hand her work as a program committee member for ICSE'26, where I served as area chair and interacted with Jialun on several occasions. She took her role seriously, providing high-quality reviews in a timely manner and contributing meaningfully to the discussions among the program committee members. Extrapolating from this experience, I am confident that Jialun will be an excellent citizen of any academic institution she joins.

While I cannot comment on her teaching skills due to my lack of experience in a classroom setting, I believe that her strong communication skills and extensive mentoring experience will enable her to be an effective teacher.

Overall, Dr. Jialun Cao is an **outstanding candidate** for a tenure-track position at a top-tier academic institution. She is a rising star, who has produced high-impact research, demonstrated strong leadership skills, and actively contributed to the academic community. I consider Jialun to be at a level comparable to the strongest tenure-track hires at CISP. Hence, **I wholeheartedly recommend her** for consideration and am confident that she will make significant contributions to any institution fortunate enough to have her on their faculty.

Sincerely,



Michael Pradel